
CF09: COMPLETE FORMALISM — GAUGE FIELD EMERGENCE VIA BERRY CONNECTION IN VDM (WEINBERG–WITTEN DEFENSE)

Justin K. Lietz 
Neuroca, Inc.
justin@neuroca.ai

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Abstract

The Void Dynamics Model (VDM) “force sector” requires long-range interactions to arise from coarse degrees of freedom without postulating fundamental gauge fields. This CF specifies a concrete U(1) construction: an emergent gauge potential is defined as the Berry connection of a low-energy spinor bundle (imported from the domain-wall sector), and its curvature is identified with the electromagnetic field strength. Under locality and gauge redundancy, the low-energy effective action admits a derivative expansion whose leading gauge-invariant term is the Maxwell operator $\int F_{\mu\nu}F^{\mu\nu}$. Two attack surfaces are treated as decisive falsifiers: (i) the connection must yield nontrivial, gauge-invariant plaquette curvature and predominantly transverse modes, and (ii) the emergent photon must remain gapless (within declared tolerance), producing a Coulomb $1/r$ potential rather than a Yukawa tail. Compatibility with Weinberg–Witten is treated operationally: the emergent A_μ is a bundle connection defined only up to local phase, not a gauge-invariant Lorentz vector created by a local operator in the same Hilbert space as the conserved current. The deliverable is a publishable derivation plus a compact gate suite (G1–G6) that a companion notebook CFN must implement with auditable artifacts.

Keywords Berry connection · quantum geometric tensor · emergent gauge field · Maxwell action · lattice gauge theory · Weinberg–Witten theorem · domain-wall fermions · Void Dynamics Model

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1 Objective, scope, and non-claims

The VDM program posits a microscopic scalar lattice (the conservative “J-limb”) whose coarse degrees of freedom must reproduce familiar long-range forces while remaining locally causal and metriplectic at the axiomatic level [Lietz, 2025a]. The force-sector requirement relevant here (Gap Module S6) is the emergence of a *massless spin-1 mediator* with U(1) gauge redundancy and an effective Maxwell operator. This CF provides a tight, falsifiable construction of an emergent electromagnetic gauge field using the Berry connection formalism imported from CF01 [Lietz, 2025b] and low-energy spinor states imported from CF08 [Lietz, 2025c].

Deliverable. The deliverable is a publishable derivation plus a decisive gate suite that a companion notebook CFN must implement: (i) extraction of a lattice U(1) connection from state overlaps, (ii) computation of gauge-invariant curvature (plaquettes/Wilson loops), (iii) identification of the leading F^2 term in the low-energy effective action, (iv) transversality and gaplessness diagnostics, and (v) a static-potential check distinguishing Coulomb from Yukawa behavior.

Scope classification. This module is *derived-limit*: it imports standard Berry-phase and lattice-gauge constructions as intermediate tools, then binds them to VDM observables and gates. It becomes axiom-relevant only insofar as the gates are passed by VDM simulations.

Non-claims (tight).

- No novelty is claimed for Berry phases, Wilson lines, or the Maxwell action; these are classical results [Berry, 1984, Provost and Vallée, 1980, Wilson, 1974, Kogut, 1979].
- This CF does not claim that VDM predicts the numerical value of the fine-structure constant α ; the emergent coupling is treated as a parameter to be measured in the companion notebook and later constrained.
- Only abelian U(1) gauge structure is treated. Non-abelian extensions (SU(2), SU(3)) require additional bundle structure and are deferred.
- Lorentz invariance is not assumed at the lattice scale; the only requirement is an auditable low-momentum isotropy window (which may be tested in a separate symmetry module).

2 Claim inventory and decisive falsifiers

This CF treats claims as contracts: each claim has a single decisive metric and a falsifier. The gates are designed to be binary or near-binary.

2.1 Primary claims

Claim: C1 (Emergent U(1) connection from state overlaps) *Decisive metric: gauge-covariance residual on links/plaquettes below tolerance.*

Given normalized low-energy states $\{|\psi_n\rangle\}$ on lattice sites, a U(1) link variable $U_{n,\mu} \in \text{U}(1)$ can be defined from nearest-neighbor overlaps. Under local phase choices $|\psi_n\rangle \mapsto e^{i\Lambda_n}|\psi_n\rangle$, the links transform covariantly and the plaquette curvature is invariant.

Claim: C2 (Berry curvature equals electromagnetic field strength) *Decisive metric: Bianchi residual and antisymmetry residual below tolerance.*

The connection induces a curvature two-form $F_{\mu\nu}$ which is gauge-invariant, antisymmetric, and satisfies the Bianchi identity. In the continuum limit, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ matches the Berry curvature of the state bundle.

Claim: C3 (Low-energy effective action has a leading Maxwell term) *Decisive metric: coefficient fit identifies a leading F^2 operator with predicted scaling.*

Under locality and gauge redundancy, the effective action for the emergent connection admits a derivative expansion whose leading gauge-invariant operator is $\int F_{\mu\nu} F^{\mu\nu}$. The companion notebook must extract this term and verify that higher-derivative corrections are subleading on a declared infrared window.

Claim: C4 (Physical photon sector: transverse and gapless) *Decisive metric: transverse fraction $f_\perp$ exceeds threshold and mass gap bound holds.*

Physical excitations of the emergent U(1) sector are transverse (two polarizations) and gapless within declared tolerance. Operationally, this implies (i) the longitudinal fraction of $\vec{A}$ is negligible in Coulomb gauge, and (ii) dispersion fits exclude a persistent mass term. A Coulomb $1/r$ static potential is a downstream corollary.

2.2 Assumption ledger

1. **A1 (Existence of low-energy spinor bundle):** the VDM matter sector produces normalized low-energy spinor-like states with a smooth dependence on spacetime/coarse coordinates (imported from CF08) [Lietz, 2025c, Kaplan, 1992, Shamir, 1993].
2. **A2 (Adiabatic/infrared window):** there exists a scale separation so that Berry geometry is meaningful and a derivative expansion for the effective action is controlled.
3. **A3 (Nonvanishing overlaps):** nearest-neighbor overlaps $\langle \psi_n | \psi_{n+\hat{\mu}} \rangle \neq 0$ hold on the declared lattice window so that link phases are well-defined.
4. **A4 (Gauge redundancy preserved under coarse-graining):** the map from microscopic states to $|\psi\rangle$ does not explicitly break local phase freedom in the effective description.
5. **A5 (Operational Weinberg–Witten stance):** the emergent A_μ remains a *connection-level* object (defined only up to phase choice) rather than a gauge-invariant Lorentz vector created by a local operator in the same Hilbert space as the conserved current.

2.3 Decisive gates (G1–G6)

The following gates are intended to be implemented in CFN9 and logged with pass/fail metadata.

Gate: G1 (Gauge covariance / plaquette invariance) *Threshold: mean plaquette phase residual under random gauge $\leq 10^{-10}$.*

Apply a random local phase field Λ_n to $|\psi_n\rangle$ and recompute $U_{n,\mu}$ and plaquette phases. The plaquette curvature must remain invariant to numerical tolerance; otherwise C1 fails.

Gate: G2 (Bianchi identity residual) *Threshold: $\|\partial_{[\lambda} F_{\mu\nu]}\|$ (or discrete analogue) $\leq 10^{-10}$.*

Compute the discrete curl-of-curl residual for F . Failure indicates that the constructed curvature is not a consistent field strength and C2 fails.

Gate: G3 (Maxwell operator dominance) *Threshold: on an IR window, S_{eff} fit requires leading F^2 term with $R^2 \geq 0.99$.*

Fit the effective action density against candidate gauge-invariant operators F^2 , $(\partial F)^2$, etc. The F^2 coefficient must be nonzero and dominant on the declared IR window; otherwise C3 fails.

Gate: G4 (Transversality) *Threshold: transverse fraction $f_\perp \geq 0.999$.*

Compute $\vec{A}(\vec{k})$ and project into transverse/longitudinal components. If $f_\perp$ fails, the emergent field is predominantly pure gauge and C4 fails.

Gate: G5 (Gaplessness / photon mass bound) *Threshold: fit $\omega^2(k) = c^2 k^2 + m_\gamma^2$ gives $m_\gamma a < 10^{-12}$.*

Extract dispersion from correlators or mode energies and fit for m_γ . Any persistent gap beyond tolerance falsifies the “massless photon” sub-claim.

Gate: G6 (Coulomb vs Yukawa) *Threshold: potential fit residual $\leq 1\%$ over $r \in [2a, 10a]$.*

Compute a static potential $V(r)$ between test charges. A pure Coulomb fit $V(r) = A/r + B$ must dominate over a Yukawa form; otherwise long-range electromagnetism does not emerge on that window.

2.4 Companion notebook outputs (minimum)

- Gauge residual logs for G1–G2 (JSON + histograms of plaquette phases).
- Operator-basis fit for G3 (coefficients with uncertainties and a declared IR window).
- Transverse/longitudinal decomposition plots and $f_\perp$ table for G4.
- Dispersion fit with m_γ estimate and confidence interval for G5.
- Coulomb fit residual plot and CSV for G6.

3 Canon registries, symbols, and units

This CF is written to be compatible with the VDM canon registry discipline: axioms [Lietz, 2025a], equation registry [Lietz, 2025d], and validation metrics [Lietz, 2025e] define the shared language. The Berry-geometry construction is imported from CF01 [Lietz, 2025b]; the low-energy spinor bundle is imported from CF08 [Lietz, 2025c]. Gauge-sector gates and artifact paths are aligned with hypothesis H006 [Lietz, 2025f].

Unless otherwise stated, natural units $\hbar = c = 1$ are used. In lattice diagnostics, the lattice spacing a is retained explicitly when it clarifies scaling; otherwise “lattice units” correspond to setting $a = 1$.

4 Foundations: Berry geometry and the low-energy state bundle

4.1 State bundle over spacetime (or coarse coordinates)

Let M denote the effective 3+1D spacetime (or a coarse-grained lattice approximation). Assume the VDM matter-sector construction yields, at each $x \in M$, a normalized low-energy state $|\psi(x)\rangle$ in an effective Hilbert space. The physical content is invariant under local phase choices

$$|\psi(x)\rangle \mapsto e^{i\Lambda(x)} |\psi(x)\rangle, \quad (1)$$

so $|\psi\rangle$ defines a $U(1)$ principal bundle (or associated complex line bundle) over M [Nakahara, 2003].

4.2 Quantum geometric tensor and its decomposition

The quantum geometric tensor (QGT) provides a coordinate-free way to extract both a metric and a curvature from $|\psi(x)\rangle$ [Provost and Vallée, 1980]. In local coordinates x^μ , define

$$Q_{\mu\nu}(x) \equiv \langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\mu \psi | \psi \rangle \langle \psi | \partial_\nu \psi \rangle. \quad (2)$$

Its real part is the Fubini–Study metric $g_{\mu\nu} = \text{Re } Q_{\mu\nu}$, and its imaginary part encodes the Berry curvature. In particular, defining

$$A_\mu(x) \equiv i \langle \psi(x) | \partial_\mu \psi(x) \rangle, \quad (3)$$

one has

$$F_{\mu\nu}(x) \equiv \partial_\mu A_\nu - \partial_\nu A_\mu = 2 \text{Im } Q_{\mu\nu}(x). \quad (4)$$

The Berry-phase origin of A is classical [Berry, 1984, Simon, 1983]; CF01 adapts (2)–(4) to VDM notation [Lietz, 2025b].

4.3 Import: emergent spinors from the domain-wall sector

CF08 imports a standard lattice mechanism (domain-wall fermions) to obtain chiral, wall-localized spinor modes [Lietz, 2025c, Kaplan, 1992, Shamir, 1993]. CF09 does not re-derive that construction; it assumes the output is a family of normalized low-energy spinor states $|\psi(x)\rangle$ with phase redundancy (1). The present contribution is to show that the bundle geometry of these states yields an emergent $U(1)$ gauge sector.

5 Berry connection as an emergent U(1) gauge potential

5.1 Continuum construction and gauge transformation

Definition 1 (Berry connection as emergent gauge potential). For normalized low-energy states $|\psi(x)\rangle$, define A_μ by (3).

Under a local phase choice (1), one finds

$$A_\mu(x) \mapsto A_\mu(x) + \partial_\mu \Lambda(x), \quad (5)$$

which is precisely the U(1) gauge transformation law. Reality follows from normalization: $A_\mu^* = A_\mu$.

5.2 Parallel transport and Wilson lines

The connection A determines parallel transport: moving the state from x to $x + dx$ accumulates a phase $e^{-iA_\mu dx^\mu}$. Along a curve γ , the holonomy is the Wilson-line phase

$$\mathcal{W}[\gamma] = \exp \left(-i \int_\gamma A_\mu dx^\mu \right), \quad (6)$$

and for a closed loop $\partial\Sigma$ it equals the flux of curvature through Σ (Stokes' theorem), linking the construction to electromagnetic Aharonov–Bohm phases [Nakahara, 2003].

5.3 Discrete lattice realization via overlaps

On a lattice with sites n and unit vectors $\hat{\mu}$, define a U(1) link variable from overlaps:

$$U_{n,\mu} \equiv \frac{\langle \psi_n | \psi_{n+\hat{\mu}} \rangle}{|\langle \psi_n | \psi_{n+\hat{\mu}} \rangle|} \in \text{U}(1). \quad (7)$$

If $|\psi_n\rangle \mapsto e^{i\Lambda_n} |\psi_n\rangle$, then

$$U_{n,\mu} \mapsto e^{i(\Lambda_{n+\hat{\mu}} - \Lambda_n)} U_{n,\mu}, \quad (8)$$

matching the standard lattice gauge transformation law [Wilson, 1974, Kogut, 1979]. A lattice gauge potential (up to branch choice) can be defined as

$$A_{n,\mu} \equiv -\frac{1}{a} \text{Arg}(U_{n,\mu}), \quad U_{n,\mu} = e^{-iaA_{n,\mu}}. \quad (9)$$

The plaquette phase gives a discrete curvature:

$$U_{n,\mu\nu} \equiv U_{n,\mu} U_{n+\hat{\mu},\nu} U_{n+\hat{\mu},\nu}^* U_{n,\mu}^*, \quad (10)$$

and

$$F_{n,\mu\nu} \equiv -\frac{1}{a^2} \text{Arg}(U_{n,\mu\nu}). \quad (11)$$

By construction, $U_{n,\mu\nu}$ (and hence $F_{n,\mu\nu}$) is gauge-invariant under (8). The continuum limit $a \rightarrow 0$ recovers $F_{\mu\nu}$ in (4).

Remark 1 (VDM-specific note). In standard lattice gauge theory, $U_{n,\mu}$ may be treated as a fundamental degree of freedom. In VDM, $U_{n,\mu}$ is a *derived observable* extracted from the emergent state bundle; gauge structure is therefore interpreted as an emergent redundancy rather than an additional postulate.

6 Field strength from Berry curvature

6.1 Curvature as electromagnetic field tensor

Definition 2 (Field strength). Define the field strength as the curvature two-form

$$F_{\mu\nu}(x) = \partial_\mu A_\nu - \partial_\nu A_\mu. \quad (12)$$

Gauge invariance is immediate from (5). Equivalently, (4) yields

$$F_{\mu\nu} = i(\langle \partial_\mu \psi | \partial_\nu \psi \rangle - \langle \partial_\nu \psi | \partial_\mu \psi \rangle), \quad (13)$$

which is manifestly invariant under (1).

6.2 Bianchi identity

Because $F = dA$, it satisfies the Bianchi identity

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0. \quad (14)$$

The discrete analogue uses oriented products of plaquettes around a cube; gate G2 is the operational form.

6.3 Electric and magnetic fields

In 3+1 notation, $E^i = F^{0i}$ and $B^i = \frac{1}{2}\epsilon^{ijk}F_{jk}$. A nonzero $\vec{B} = \nabla \times \vec{A}$ is the simplest “smoking gun” that $\vec{A}$ is not a pure gradient.

7 Low-energy effective action and the Maxwell operator

7.1 Why F^2 is the leading gauge-invariant operator

The low-energy effective action $S_{\text{eff}}[A]$ is assumed to be local on the coarse scale (a VDM axiom-level discipline) and invariant under (5). These constraints imply that S_{eff} depends on A only through F (and its derivatives) and admits a derivative expansion

$$S_{\text{eff}}[A] = \int d^4x \left[-\frac{1}{4g^2} F_{\mu\nu}F^{\mu\nu} + c_1 (\partial_\alpha F_{\mu\nu})(\partial^\alpha F^{\mu\nu}) + \dots \right]. \quad (15)$$

The first term is the Maxwell operator; higher terms are suppressed on an infrared window where $\partial \ll \Lambda$ for some cutoff $\Lambda \sim \pi/a$. Gate G3 is designed to test that F^2 dominates on a declared window.

7.2 Connection to the quantum geometric tensor

Within Berry geometry, F arises as the antisymmetric part of the QGT (4). A minimal scaling estimate for the emergent coupling may be obtained by dimensional analysis: if the gauge field is induced by a localized domain-wall profile $\chi_0(z)$ (normalized zero mode in the auxiliary coordinate), then a typical coefficient scales like

$$\frac{1}{g^2} \propto \frac{1}{a^2} \int dz |\chi_0(z)|^4, \quad (16)$$

up to model-dependent constants. The companion notebook should treat (16) as a *scaling hypothesis* and verify coefficient scaling empirically rather than asserting it.

8 Transversality, gaplessness, and long-range force

8.1 Transversality of physical modes

Any vector field decomposes (Helmholtz) into transverse and longitudinal components, $\vec{A} = \vec{A}_\perp + \nabla\chi$ with $\nabla \cdot \vec{A}_\perp = 0$. Gauge transformations shift only the longitudinal piece, so physical photon modes reside in $\vec{A}_\perp$. In Fourier space, $\vec{A}_\perp(\vec{k})$ is obtained by projecting orthogonally to $\vec{k}$.

Theorem 1 (Nontrivial curvature implies non-pure-gauge content). *If $A_\mu = \partial_\mu\phi$ globally (pure gauge), then $F_{\mu\nu} = 0$. Therefore any nontrivial curvature measured by plaquette phases implies that A is not globally a gradient.*

Gate G4 quantifies this by requiring a large transverse fraction on the declared momentum window.

8.2 Masslessness as a measurable gap gate

Gauge redundancy forbids a Proca mass term $m^2 A_\mu A^\mu$ in a gauge-invariant effective action. However, coarse-graining and discretization can break gauge redundancy softly; therefore “masslessness” is treated operationally as a dispersion gate (G5). For context, experimental bounds constrain the photon mass to be extremely small [Workman et al., 2022].

8.3 Coulomb potential

In the static limit, Maxwell theory implies $\nabla \cdot \vec{E} = 4\pi\rho$ and $\vec{E} = -\nabla\phi$. A point charge yields

$$\phi(\vec{x}) = \frac{q}{4\pi|\vec{x}|}, \quad V(r) \propto \frac{1}{r}. \quad (17)$$

A Yukawa form e^{-mr}/r indicates a mass gap. Gate G6 distinguishes these cases on a finite lattice window.

9 Weinberg–Witten compatibility as an operational gate

9.1 The theorem (context)

Weinberg and Witten showed that in a Lorentz-invariant quantum field theory admitting a Lorentz-covariant conserved current, massless particles of spin $j > \frac{1}{2}$ cannot carry the associated charge; similarly, with a Lorentz-covariant conserved stress tensor, massless particles of spin $j > 1$ cannot carry energy–momentum in the required sense [Weinberg and Witten, 1980]. The theorem is often invoked as a “no-go” for composite gauge bosons.

9.2 Why the Berry-connection construction targets the loophole

The present construction does not attempt to build a gauge boson as a gauge-invariant local composite operator. Instead:

- A_μ is defined as a *connection* on a bundle of states (3). It is only defined up to (5) and is therefore not itself a gauge-invariant local observable.
- Gauge-invariant observables are curvature- or holonomy-based (plaquettes/Wilson loops), not the connection components.
- The effective conserved current (e.g., $\bar{\psi}\gamma^\mu\psi$) lives in the emergent fermion sector; the gauge field is geometric data controlling parallel transport of those states.

This matches the standard geometric formulation of gauge theory [Nakahara, 2003]. Operationally, the “Weinberg–Witten defense” is therefore treated as an *implementation constraint*: CFN9 must demonstrate that the emergent U(1) field appears as bundle connection data extracted from overlaps and that physical predictions depend only on gauge-invariant holonomies. If instead CFN produces a gauge-invariant Lorentz vector operator that creates the would-be photon state in the same Hilbert space as the conserved current, the intended loophole is closed and the model must be reconsidered.

10 Constructive algorithms and validation protocol (for CFN9)

This section specifies an implementation plan without committing to any particular software framework.

10.1 Algorithm A: Build links and curvature from states

1. Input: normalized states $|\psi_n\rangle$ on sites n .
2. Compute overlaps $o_{n,\mu} = \langle\psi_n | \psi_{n+\hat{\mu}}\rangle$.
3. Define links $U_{n,\mu} = o_{n,\mu}/|o_{n,\mu}|$ (7).
4. Compute plaquettes $U_{n,\mu\nu}$ (10) and curvature $F_{n,\mu\nu}$ (11).

10.2 Algorithm B: Gauge-invariance stress test (G1)

1. Sample random phases Λ_n and transform $|\psi_n\rangle \mapsto e^{i\Lambda_n}|\psi_n\rangle$.
2. Recompute $U'_{n,\mu}$ and $U'_{n,\mu\nu}$.
3. Compare plaquette phases: $\Delta_{n,\mu\nu} = \text{Arg}(U'_{n,\mu\nu}) - \text{Arg}(U_{n,\mu\nu})$ modulo 2π .
4. Gate: mean and max $|\Delta|$ below tolerance.

10.3 Algorithm C: Operator-basis fit for the effective action (G3)

1. Construct candidate local operators: $\mathcal{O}_0 = F^2$, $\mathcal{O}_1 = (\partial F)^2$, $\dots$.
2. Fit action density (or log-likelihood) to $\sum_i c_i \mathcal{O}_i$ on a declared IR window.
3. Gate: $c_0 \neq 0$ and dominates; fit quality meets threshold.

10.4 Algorithm D: Transverse projection and dispersion (G4–G5)

1. Fourier transform $\vec{A}(\vec{x}) \rightarrow \vec{A}(\vec{k})$.
2. Project $\vec{A}_\perp = \vec{A} - (\hat{k} \cdot \vec{A})\hat{k}$.
3. Compute $f_\perp = \sum_{\vec{k}} \|\vec{A}_\perp\|^2 / \sum_{\vec{k}} \|\vec{A}\|^2$.
4. Extract dispersion $\omega(k)$ from correlators and fit $\omega^2 = c^2 k^2 + m_\gamma^2$.

10.5 Algorithm E: Static potential (G6)

1. Place test charges or impose a source ρ .
2. Solve discrete Poisson/Laplace equation for ϕ or compute Wilson loops.
3. Fit $V(r)$ to Coulomb and Yukawa forms; compute residuals.

11 Integration map

- **CF01** $\rightarrow$ **CF09**: Berry connection and QGT supply the geometric objects A and F [Lietz, 2025b].
- **CF08** $\rightarrow$ **CF09**: domain-wall localized spinor states supply the bundle sections $|\psi\rangle$ from which A is constructed [Lietz, 2025c].
- **CF09** $\rightarrow$ **later modules**: once gates pass, A provides the force-sector ingredient needed for electromagnetic coupling, transport, and (eventually) non-abelian generalizations.

12 Limitations and future work

- **Non-abelian gauge groups**: extending overlaps to an $SU(N)$ connection requires degeneracy structure and a non-abelian Berry connection.
- **Running coupling and quantum corrections**: the present CF only specifies the leading Maxwell operator and does not derive renormalization-group running.
- **Backreaction**: coupling between the emergent gauge sector and the underlying scalar lattice dynamics is not modeled here; it must be specified in a combined dynamics module.
- **Weinberg–Witten risk**: if the implementation yields a gauge-invariant local vector operator creating the photon state in the same Hilbert space as the conserved current, the “connection loophole” is not available.

13 Provenance

This build is intended for Zenodo-style archival.

- Canon snapshot commit (from provenance manifest): 91332f4cf89a52d239eaab6659b8df8d8b9f3202.
- Source basis: Derivation/Complete-Formalisms/CF09_Gauge_Emergence_Berry_Connection.md (repository snapshot; LaTeX content adapted for publication).
- Companion hypothesis alignment: H006 gates and artifact paths [Lietz, 2025f].

End of CF09.

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